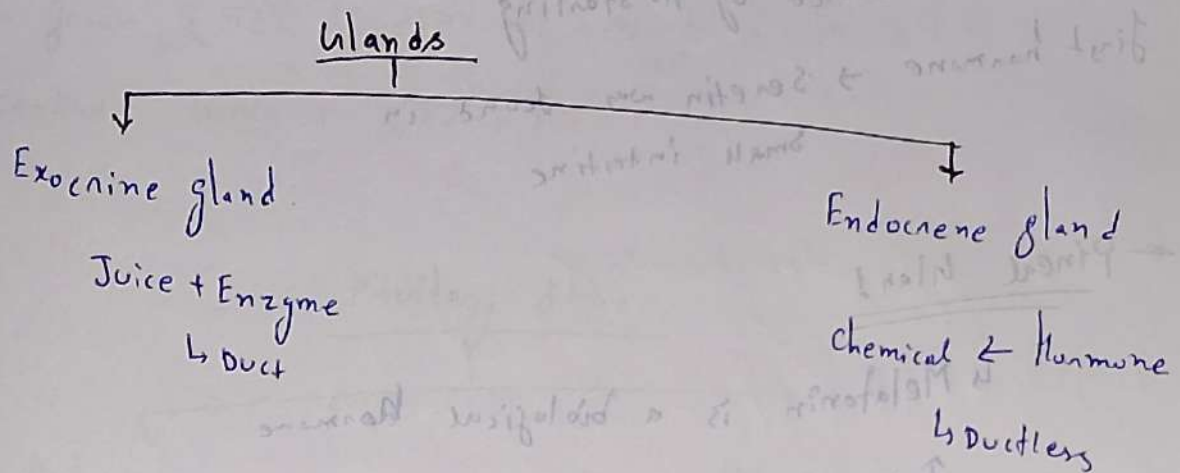


Glands



Exocrine gland

- i. Lacrimal gland → Eyes → Tears → Smallest gland
 - ii. Liver → Bile juice / Largest gland
 - iii. Mammary gland → Milk formation
 - iv. Gastric gland → Gastric juice (HCl)
 - v. Salivary gland → Saliva → mouth
 - vi. Sweat glands / Sebaceous gland → skin
 - vii. Intestinal gland → Small intestine
- Largest Exocrine → Liver
- Smallest Exo → Tear gland

Endocrine gland

- | | |
|--------------------|---------------|
| i) Hypothalamus | vii) Thyroid |
| ii) Thyroid | viii) Ovaries |
| iii) Adrenal gland | ix) Testes |
| iv) Pineal gland | |
| v) Pituitary gland | |

• Endocrine gland → Hormone

↳ 1902

↳ discovered by H. Starling

First hormone → Secretin was found in
Small intestine

• Pineal gland

↳ Melatonin is a biological hormone

↳ helps in sleeping at night

wakes in the morning

• Biological clock gland

- Diurnal cycle

- Night hormone

• Melanin

↳ Skin color

↳ it present in high amount in
skin (Dark/Black)

↳ it low in body

(Fair / Light-skinned)

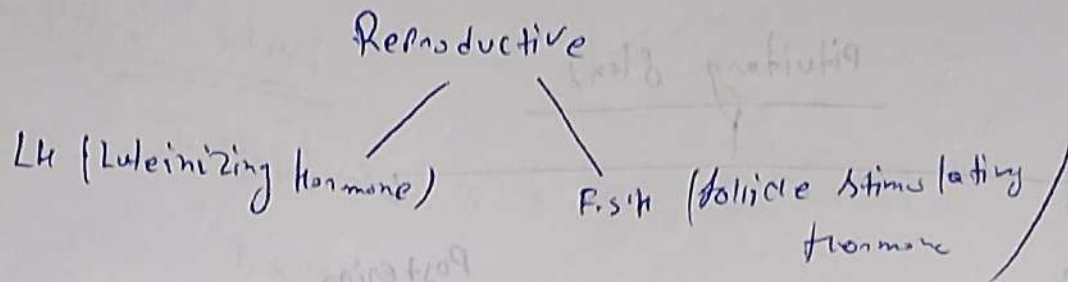
Low of melanin also cause a
disease name Insomnia

→
जीने में आने की बीमारी

• GnRH:

• Gonadotropin Releasing Hormone

- Reproductive organ growth in both male & female



• ~~Posterior Pituitary gland~~

* Vasopressin / ADH :- Anti-diuretic Hormone

- controls water in body

Urin → water.

* Oxytocin

- found in female

- Labour Pain (helps to hold the pain at the time of delivery)

It is present in boy

it is known as Love hormone

↳ Boys

↳ Bisexual

* Hypothalamus

↳ Mom gland.

↳ master of master gland.

↳ Pituitary gland से निगल है

↳ Dopamine

↳ ~~just~~ also known as happy hormone

↳ also known as Hobby hormone

↳ Addiction hormone (जित जितने दवा Hormone)

* Thyroid gland disease

→ release Thyroxine (98%)

→ Calcitonine (2%)

Thyroxine

if Present in Low amount

• Cretinism

• 90% Live

• mentally retarded

if Present in high amount

• Grave's Disease

• Eyeballs start protruding

• BP high

Calcitonine

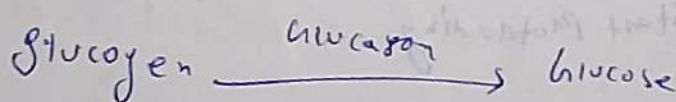
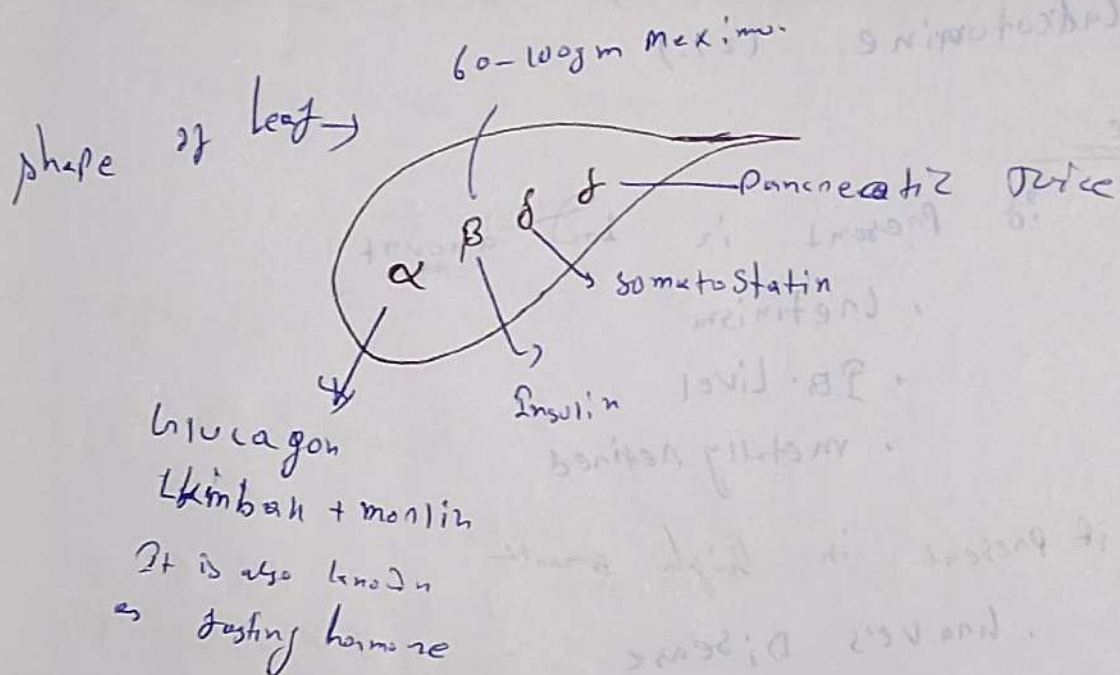
Takes calcium from Blood to Bones

* Parathyroid gland :

- Parathyroxine Hormone
- Blood
- Calcium in Bones से Blood में ले जाता है

- Pancreas : (mixed gland)
 - Juice
 - Enzyme
 - Hormone
- ↳ Islets of Langerhans
 - ↳ β -cells
 - ↳ Insulin hormone

Blood (Sugar) (Balance) Insulin



Glucose $\xrightarrow{\text{Insulin}}$ Glycogen $\rightarrow$ Liver.

Adipose tissue.

Glycogen $\xrightarrow{\text{Glucagon}}$ Glucose

* Insulin

• discovered Banting & Best

Blood glucose $\xrightarrow{\text{Insulin}}$ Glycogen

β cells $\rightarrow$ Islets of Langerhans

$\rightarrow$ Zinc help

Low production of insulin $\downarrow$ Diabetes

Type 1

• Juvenile

• After birth

• 12-13 Age

Type 2.

• Life style

• Diet

Sugar level

Fasting sugar $\rightarrow$ 70-80 mg/dL

After meal $\rightarrow$ 150-170 mg/dL

200+ then you have sugar on diabetes

• Thymus gland

↳ WBC
↳ B-lymphocyte

• maturity

• Age 22 $\frac{2}{3}$ + $\frac{1}{3}$ at 12

• gland of old age

• Thymosine

• Cortex

• cortisol

characteristics

↓
Sugar level

Stress control

↳ Aldosterone

↓
Salt

• Androgen

↓
Sexual characteristics

It is also known as 4's gland.

• Medulla

↳ Adrenaline

↳ 3F hormone (fear, fight, flight)

↳ also known as

Emergency Hormone

* Gonads gland —

releaser

• Testes — Testosterone

↳ Sperm cell

• ovaries → Progesterone

↳ pregnancy

↳ female

* LH → Luteinising Hormone

FSH → follicle stimulating hormone

Key Glands

• Largest gland — Liver

• Smallest Exocrine gland — Tear gland

• Largest Endocrine — Thyroid

11/02

• Smallest " — Pineal

• Second " — Pituitary

• H-shaped gland = Thyroid

• Emergency gland / Do or die / 3rd / 4th gland

↳ Adrenal gland

• Master gland — Pituitary

• Mom gland → Hypothalamus

also known as Thermometer gland.

Age 1st → Thymus.